EE3404 Circuits I - Full Solutions Study Guide

Quiz 1: Can a Capacitor Voltage Jump?

Theory:
An ideal capacitor obeys i = C(dv/dt). Therefore its voltage cannot change instantaneously because that would require infinite current.

Given:
C = 100 µF = 1×10^-4 F
vC(0-) = 6 V, claimed vC(0+) = 0 V.

Solution:
1. The claim is incorrect. For an ideal capacitor, vC(0+) = vC(0-) = 6 V.
2. Using i = C(dv/dt), an instantaneous voltage jump means dt→0, so dv/dt→∞ and therefore i→∞. Infinite current is not physically possible.
3. If the 6 V drop occurs uniformly in 1 µs:
i = CΔV/Δt = (100×10^-6)(-6)/(1×10^-6) = -600 A. Magnitude = 600 A.
4. Even an extremely fast voltage change requires a very large current. A true instantaneous jump would require infinite current.
5. Real capacitors may appear to change voltage rapidly because of ESR, parasitic inductance, dielectric breakdown or measurement bandwidth limitations.

Exam Tips:
• Always convert µF to F and µs to s.
• State capacitor voltage continuity.
• Quote i=C(dv/dt).
• Explain the physical meaning, not just the mathematics.

Common Viva Questions:
Why can't capacitor voltage jump? Because infinite current would be required.
What quantity is continuous? Capacitor voltage.
What does a capacitor store? Energy in an electric field.